

Algorithmically-Optimized Real-Time 18 Tb/s Throughput over a 16,608 km Trans-Pacific Subsea Link

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Abstract We report 18 Tb/s throughput across a 16,608 km trans-Pacific subsea link using an algorithmically-optimized channel plan, for a record 298.9 Pb/s·km. Bidirectional SNR margins were 0.18–0.75 dB for all 28 channels. Additionally, an 800 Gb/s (173 GBd) signal reached 4.57 bits/s/Hz spectral efficiency. © 2026 The Author(s)

Introduction

The expansion of global cloud computing, 4k/8k UHD streaming, and AI-driven data exchange has placed unprecedented demand on international subsea links. As primary arteries of connectivity, subsea cables must support large inter-continental traffic over extreme distances. Technical innovation to improve throughput focuses on improving modem spectral efficiency via advanced modulation, nonlinear compensation and coding, whilst reducing power consumption through higher symbol rate operation.

Beyond standard ASE noise and Kerr-induced nonlinearity, ultra-long-haul regimes encounter complex temporal and phase distortions that degrade the receiver's Total Signal-to-Noise Ratio (SNR_{Tot}) [1,2]. Over trans-Pacific distances, these degradations can lead to highly variable performance across the spectrum that challenges the limits of traditional, uniform channel planning. In this context, software-defined optimization presents a powerful opportunity to unlock untapped capacity. Recursive optimization routines offer a systematic way to navigate the multi-dimensional solution space provided by modern transceivers—intelligently balancing granular symbol rates, probabilistic constellation shaping (PCS), and line rate tunability to maximize throughput while maintaining robust operating margins.

This paper demonstrates the efficacy of such

an approach through the characterization of a fiber-pair on the 16,608 km trans-Pacific Bifrost cable. Central to this achievement is the implementation of a recursive optimization routine designed to autonomously predict and implement a maximum aggregate throughput plan. By targeting a deployable 0.2 dB ESNR operating margin, we show how algorithmic optimization can extract untapped capacity. Using this method, an 18 Tb/s channel plan was successfully operated for 8 hours with no post-FEC erroring, achieving what we believe to be a record single-core, C-band throughput-distance product of 298.9 Pb/s·km.

Bifrost Cable and Equipment Setup

The Bifrost cable is shown schematically in Fig. 1. (a). The main trunk of the cable spans ~16,584 km between the Cable Landing Station (CLS) sites at Tuas, Singapore and Grover Beach, USA. Additional terrestrial backhaul sections were used to extend the system to Point of Presence (PoP) sites at Tuas (0.8 km) and San Luis Obispo (SLO, 23 km), respectively. Total transmission length was 16,608 km. The submarine line terminal equipment (SLTE) was situated in the PoP sites and is shown in Fig. 1. (b). It comprised fourteen commercially-available real-time muxponders per end, each capable of transmitting and receiving two modulated signals, for a maximum channel count of 28. The line rate of

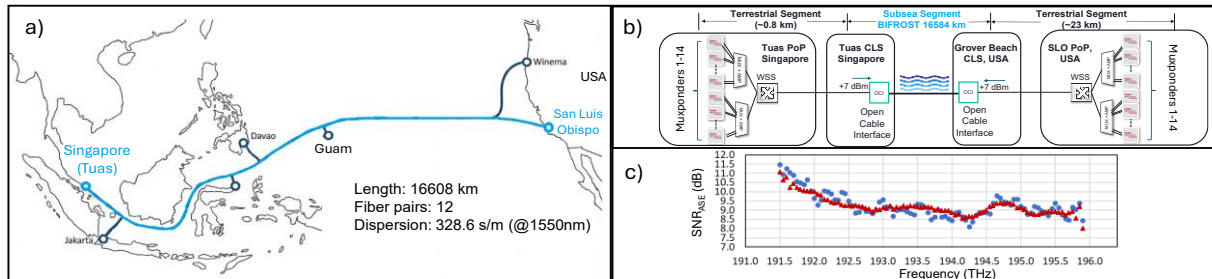


Fig. 1. (a) Schematic showing 16608 km Bifrost link route (blue) (b) Schematic overview of transmission set-up (c) Measured SNR_{ASE} spectral profile for Singapore Rx (blue dots) and San Luis Obispo Rx (brown triangles).

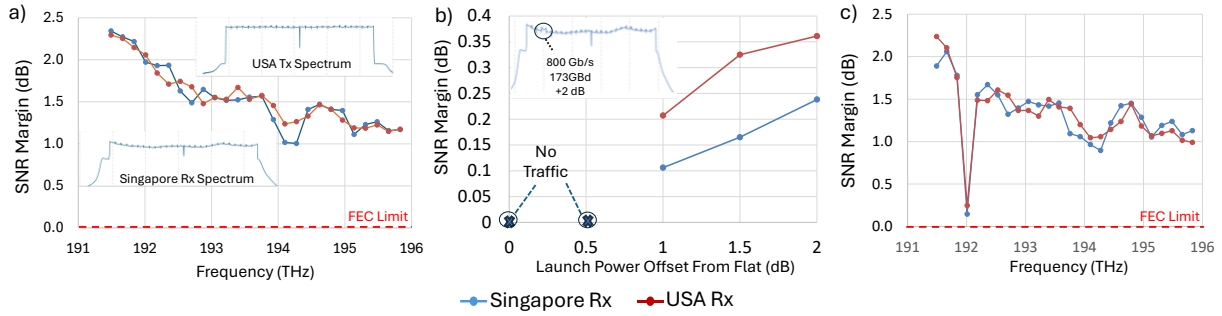


Fig. 2. (a) SNR Margin vs frequency for 26 x 600 Gb/s 170 GBd channels after 12-hour stability test. Inset are USA Tx and Singapore Rx spectra (b) Instantaneous SNR margin vs launch power offset for the 800 Gb/s 173 GBd channel with inset Singapore Rx spectrum (c) Min. SNR Margin vs Frequency for 1 x 800 Gb/s channel and 25 x 600 Gb/s channels after 12-hour stability test

each channel was adjustable in 100 Gb/s increments between 200 Gb/s and 1600 Gb/s with the ability to tune spectral efficiency using PCS. The PCS for Bifrost was based on 16 QAM. The symbol rate was tuneable between 140.1 - 200 GBd at 0.1 GBd granularity. Each channel comprised eight FDM subcarriers, and forward error correction (FEC) was throughput-optimized in terms of FEC rate according to the spectral efficiency being transmitted. Each muxponder had integrated test-set functionality with performance-monitoring capable of measuring real-time SNR_{Tot} [3], against a Required SNR (RSNR) threshold. RSNR is the SNR_{Tot} at the FEC limit for that line rate / symbol rate combination. In this paper we report SNR Margin as a figure of merit which is $(\text{SNR}_{\text{Tot}} - \text{RSNR})$ for the channel being measured. Ethernet framing was used for all channels. The 28 channels were split into two Media Channel (MC) groups comprising 14 channels each. Each 14-channel group was passively multiplexed, amplified and passed through an amplified wavelength selective switch (WSS) before connection to the cable wetplant via an Open Cable Interface (OCI) [2]. For reference purposes, the SNR_{ASE} of the link was also measured across the spectrum using a 45 x 100GHz flat-launch ASE comb and the results are shown for each direction in Fig. 1. (c).

Initial Performance Sweep

The system was set up with an initial uniform channel plan of 26 x 600 Gb/s 170 GBd channels (15.6 Tb/s total throughput) using 173 GHz spacing and a flat launch total power of +7 dBm into the OCI. The minimum SNR Margin for both directions measured after a 12-hour stability test is shown in Fig. 2. (a), with the Tx and Rx optical spectra for SLO>Singapore also shown inset. The SNR Margin reduces with increasing frequency from a maximum of 2.3 dB at 191.49125 THz. The lowest margin of 1.0 dB is seen in the SLO>Tuas direction for the channel at centre frequency 194.27275 THz. The variation of SNR Margin across the spectrum is attributed primarily

to the SNR_{ASE} profile of Fig. 1. (b).

800Gb/s transmission

To investigate the maximum line rate achievable on this system, the line rate and symbol rate of the channel at 192.01205 THz were optimized. This channel frequency was chosen to maximize SNR_{Tot} while limiting the chromatic dispersion compensation required at higher bauds. Successful transmission was ultimately achieved in both directions for 800 Gb/s 173 GBd and 340,721 ps/nm measured dispersion. It should be noted that a launch power offset of at least +1 dB from flat was required to achieve error free transmission, as shown in Fig. 2. (b). Further increasing the power offset of the 800 Gb/s channel improved the margin with no significant nonlinear roll-off observed. The system was subsequently stability tested for a further 12 hours with the 800 Gb/s channel transmitting at +2 dB power offset from flat. The minimum SNR Margin for the 800 Gb/s channel after this period is 0.146dB as shown in Fig. 2. (c).

Algorithmic Throughput Optimization

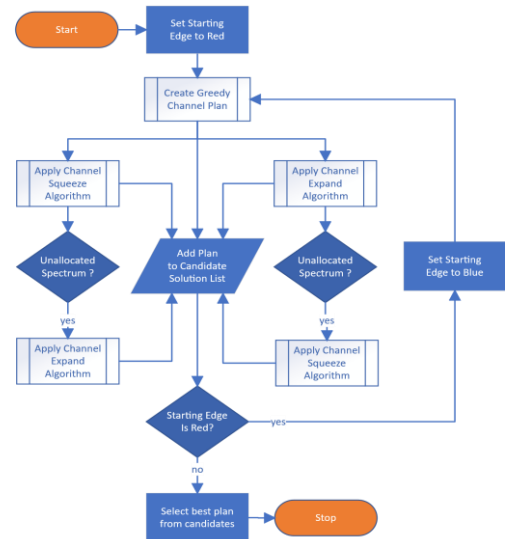


Fig. 3: Flow diagram of the recursive algorithm used for throughput optimization

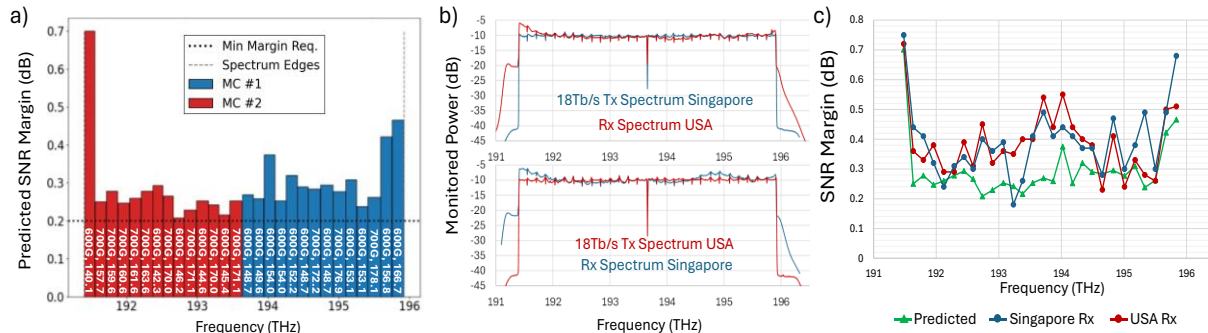


Fig. 4. (a) Algorithmically predicted 18Tb/s channel plan for a 0.2 dB minimum SNR margin target requirement. White text shows line rate in Gb/s and symbol rate in GBd (b) 18 Tb/s transmit (Tx) and receive (Rx) spectra in each direction (c) Predicted/Measured SNR margin vs Frequency per direction.

To maximize throughput for a minimum SNR margin target, a bidirectional recursive algorithm explores the solution space by initiating search processes from both spectral boundaries as shown in Fig. 3. This approach generates a candidate set of channel plan solutions, which are subsequently ranked to identify the configuration that yields the optimal throughput.

A "Greedy" algorithm first establishes a base-line channel plan via "tight packing," maximizing local spectral efficiency by minimizing channel spacing. To recover residual spectrum left by baud rate constraints, iterative "Squeeze" and "Expand" sub-algorithms are deployed to selectively decrease or increase channel bandwidths, respectively. All resulting candidate plans are cached and ranked to identify the configuration that maximizes global capacity.

For the present work, an SNR margin of 0.2 dB was targeted, with the algorithm predicting a maximum throughput of 18 Tb/s using 28 channels. The generated channel plan employed symbol rates in the range of 140.1 - 178.1 GBd and line rates of 600 Gb/s or 700 Gb/s as shown in Fig. 4. (a). A deployable guardband of 3 GHz was included between each channel as one of the constraining parameters of the algorithm. In Fig. 4. (a), the heights of the bars reflect the predicted SNR margin of each channel against the 0.2 dB target. For reference, the algorithm additionally predicted throughputs of 18.2 Tb/s (0.1 dB SNR margin) and 18.5 Tb/s (0 dB SNR margin), both with the 3 GHz guardband maintained.

Each of the 28 muxponder channels were configured to the line rates, symbol rates and centre frequencies of the 18 Tb/s channel plan and launched flat from both ends of the link. The bidirectional Tx and Rx spectra are shown in Fig. 4 (b). Error free performance was observed for each channel and direction over an 8-hour stability test. The achieved throughput-distance product was 298.9 Pb/s·km which we believe to be a record for a trans-oceanic single-core C-band submarine fiber pair.

Fig. 4 (c) plots the measured SNR margin for

each channel and direction against the algorithm-predicted values. The measurements meet the 0.2 dB target for all but one channel (0.17 dB), and the excess margin against the target ranges between 0.01 dB - 0.22 dB, excluding the first and last channels. The close agreement between measured and predicted margins is attributed to the algorithm's explicit consideration of baud-dependent variations in crosstalk penalties and implementation noise during the optimization process. The notable excess margin in the first and final channels arises from the algorithm's stop criteria prioritizing throughput as the primary objective while minimizing execution speed. While further optimization could achieve margin equalization, this was not the primary focus of the present study. This performance demonstrates the capability of an autonomous algorithm to maintain strict margin targets while maximizing delivered throughput across diverse channel conditions and a broad range of constraining parameters.

Conclusions

The efficacy of granular symbol and line rate tunability in ultra long-haul systems has been demonstrated on the 16,608 km Bifrost submarine cable in conjunction with algorithmic-optimization of the channel plan. We validated an algorithmic approach to optimizing throughput, achieving 18 Tb/s from 28 channels with zero post-FEC errors for what we believe to be a record real-time 298.9 Pb/s·km throughput-distance product on a C-band only single core subsea link. Operating SNR margins of 0.17-0.75 dB were measured indicating that this channel plan could potentially be deployed.

We additionally demonstrated the transmission of a real-time 800 Gb/s, 173 GBd signal over this distance at 4.57 bits/s/Hz spectral efficiency. We believe that these results mark a significant advancement in overall transmission performance for real-time trans-Pacific communication.

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